Arista MG-ACC-70 Node Failure SOP for ASIC Memory Parity Errors and Line Card Crashes

Node: MG-ACC-70 | **Vendor:** Arista | **Severity:** Minor | **Alarm Code:** ALARM_7888

Description

This Standard Operating Procedure (SOP) is designed to address hardware/environmental faults resulting in ASIC memory parity errors and line card crashes on the Arista MG-ACC-70 node in the MG Road Core site. The MG-ACC-70 node operates as an Access role within the Arista network architecture, utilizing Arista's Extensible Operating System (EOS) to manage data plane operations. The failure domain is attributed to the loss of power supply redundancy, which can cause the ASIC to fail, leading to line card crashes and ultimately, ASIC memory parity errors. The Arista EOS, utilizing its advanced hardware and software features, such as the Arista 7500R series, provides a robust framework for managing network operations, including fault detection and isolation. The EOS's ability to monitor and respond to hardware faults, such as ASIC memory parity errors, ensures network stability and uptime. However, in cases where the power supply redundancy is lost, the EOS's ability to detect and respond to these faults is impaired, leading to potential network disruptions.

Pre-requisite Checks

- Verify the site location is MG Road Core and the node ID is MG-ACC-70.

- Confirm the role of the node is Access and the vendor is Arista.

- Check the alarm code for ALARM_7888 and severity level for Minor.

- Ensure the redundant link status is verified and functional.

- Dispatch remote hands to MG Road Core to physically inspect the node and perform necessary checks.

Potential Root Causes

Cause	Probability
Power Supply Redundancy Loss	High
ASIC Memory Parity Errors	Medium
Line Card Crashes	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware ASIC status, Expected Output: ASIC Status: OK, No Errors, Description: This command is used to check the ASIC status and verify if there are any errors. The output will display the ASIC status and any error messages.
```

```
Command: show diagnostic result, Expected Output: Diagnostic Result: OK, No Errors, Description: This command is used to check the diagnostic result and verify if there are any errors. The output will display the diagnostic result and any error messages.

Command: show log, Expected Output: Log Messages: No Errors, Description: This command is used to check the log messages and verify if there are any errors. The output will display the log messages and any error messages.
```

Resolution Steps

Step 1: Step: Power Cycle the Node, Command: reload, Description: This step is used to power cycle the node and reset the ASIC. The reload command will power cycle the node and restart the EOS.

Step 2: Step: Replace the Power Supply, Command: replace power-supply, Description: This step is used to replace the power supply and ensure redundancy. The replace power-supply command will replace the power supply and ensure the node is operational.

Step 3: Step: Replace the ASIC, Command: replace ASIC, Description: This step is used to replace the ASIC and ensure the node is operational. The replace ASIC command will replace the ASIC and ensure the node is operational.

Post-Check Verification

```
Command: show hardware ASIC status, Expected Output: ASIC Status: OK, No Errors, Description: This command is used to verify the ASIC status and ensure there are no errors. The output will display the ASIC status and any error messages.

Command: show diagnostic result, Expected Output: Diagnostic Result: OK, No Errors, Description: This command is used to verify the diagnostic result and ensure there are no errors. The output will display the diagnostic result and any error messages.

Command: show log, Expected Output: Log Messages: No Errors, Description: This command is used to verify the log messages and ensure there are no errors. The output will display the log messages and any error messages.
```

Roll-back Procedures

- Step: Restore the Original Configuration, Command: configure replace, Description: This step is used to restore the original configuration and ensure the node is operational. The configure replace command will restore the original configuration and ensure the node is operational.

- Step: Replace the Power Supply, Command: replace power-supply, Description: This step is used to replace the power supply and ensure redundancy. The replace power-supply command will replace the power supply and ensure the node is operational.

- Step: Replace the ASIC, Command: replace ASIC, Description: This step is used to replace the ASIC and ensure the node is operational. The replace ASIC command will replace the ASIC and ensure the node is operational.